CEREBRO-X

Computational Drug-DDS Engineering Report

Field	Value
Drug Name	Nusinersen
Molecular Class	oligonucleotide
Molecular Weight	5737.6 Da
Recommended DDS	ApoE-LNP
Carrier Type	lnp
Surface Ligand	ApoE
BBB Enhancement	100.0%
Composite Score	78.1/100
Disease Indication	CNS
Report Generated	2026-09-05 18:07 UTC
Created by	Muhammad Talaat | CEREBRO-X
Modules Completed	50/62 science modules

■ *This report is generated from computational models. All predictions must be validated experimentally before clinical application.*

1. Drug Molecular Profile

Property	Value	Source / Method
Name	Nusinersen	Input
MW (Da)	5737.6	ChEMBL / UniProt
LogP	N/A	ADMET predictor
Half-life (days)	N/A	PubMed NLP
Protein Binding (%)	N/A	Calculated
BBB Penetration (%)	N/A	BBB model
Molecule Class	oligonucleotide	Input / inferred
Aqueous Solubility	N/A	ADMET
P-gp Substrate	Unknown	QSAR panel
ADMET Score	N/A	Multi-endpoint
ADMET Grade	N/A	A-F scale

1a. Drug Delivery Barriers Identified (Auto-detected)

The pipeline automatically identified 2 delivery barriers based on molecular data. Each is resolved by the recommended DDS.

[HIGH] Poor BBB penetration	
Evidence:	Native BBB permeability = 5.0% (threshold: >10%)
Without DDS:	Only 5.0% reaches brain -- insufficient for therapeutic effect
With DDS:	100.0% BBB crossing -- 20.0x improvement
DDS Solution:	lnp with ApoE surface targeting
Scientific basis:	Receptor-mediated transcytosis via ApoE increases BBB penetration from 5.0% to 100.0% (+95.0%). Carrier bypasses passive

[CRITICAL] Large molecular weight -- passive BBB diffusion impossible	
Evidence:	MW = 5.7 kDa (Lipinski Rule 5 limit: 500 Da for passive)
Without DDS:	0% brain penetration -- purely physical barrier
With DDS:	Active transcytosis delivers 10.0% payload to CNS
DDS Solution:	lnp encapsulation (75% EE)
Scientific basis:	BBB tight junctions (pore ~1nm) physically block molecules >500Da. Encapsulation in lnp (80nm) enables receptor-mediated

2. DDS Ranking — Composite Score Methodology

All DDS formulations ranked by Composite Score = weighted combination of DLVO colloidal stability, transcytosis driving force, endosomal escape, stealth index, BBB receptor affinity, and CNS bioavailability. Scores are physics-derived — not estimated.

DDS Name	Carrier	Ligand	Score	DLVO kT
ApoE-LNP	lnp	ApoE	78.14	-6.24
AAV9-vector	aav9	None	65.94	-2.37
RVG29-Liposome-pH	liposome	RVG29	54.90	14.84
Tf-PEG-Liposome	liposome	Transferrin	54.36	8.90
RVG29-PLGA-NP	plga	RVG29	40.64	5.83
Tf-SLN-Stealth	solid_lipid	Transferrin	39.47	-0.12
Lf-Polymer-Micelle	micelle	Lactoferrin	38.31	-4.06
Bare-PLGA	plga	None	36.75	-16.16

2a. Recommended DDS — Detailed Profile

Parameter	Value	Biophysical Meaning
Composite Score	78.1	Overall Drug+DDS suitability (0-100)
BBB Enhancement	100.0%	% of dose crossing BBB via receptor transcytosis
CNS Bioavailability	44.0%	% dose reaching brain as free drug
DLVO Stability	-6.2 kT	>25kT = colloidally stable in blood
Endosomal Escape	0.52	Fraction escaping lysosomes (0-1)
Stealth Index	1.00	MPS evasion (0=opsonised, 1=invisible)
Protein Corona	N/A nm	Corona thickness; thicker = faster clearance
MPS Clearance	N/A h	Hours before liver/spleen removes DDS
Payload Efficiency	N/A%	% dose delivered as active drug to CNS
CARPA Risk	0.20	Complement activation risk (0=safe)

3. PBPK-CNS Digital Twin — 6-Compartment Simulation

Physiologically Based Pharmacokinetic model solved with scipy Radau ODE integrator (stiff system). Compartments: Plasma, BBB endothelium, Brain ISF, Brain cells, CSF, Peripheral. All parameters from molecular data — not estimated.

PK Metric	Value	Compartment	Source
Model	BiologicPBPK (FcRn + CNS transcytosis) | T½=336.0h	—	
Cmax (brain)	0.006286 µg/mL	Brain/CNS	Peak brain concentration
AUC (brain)	1.36894 µg·h/mL	Brain/CNS	Total brain exposure
AUC (plasma)	622.320 µg·h/mL	Plasma	Systemic exposure
Kp,brain	0.00220	Brain/Plasma	Partition coefficient
BBB penetration	0.2200%	BBB	BiologicPBPK (two-compartment + FcRn + CNS transcy
Kp,uu brain	N/A	Unbound	Unbound Kp (pharmacodynamically relevant)
t_above 10%Cmax	N/A h	Brain ISF	Duration above therapeutic threshold
Glymphatic t1/2	N/A h	Brain	Clearance half-life in brain
BBB PS_in	N/A mL/h	BBB	Permeability-surface area product
BBB PS_out	N/A mL/h	BBB	Efflux transport rate
Disease state	healthy	BBB	BBB integrity modifier applied

4. In-Silico Drug Release Profile

Release model: First Order | Release order: Sustained first-order | Max EE: 75%

Metric	Blood (pH 7.4)	Endosomal (pH 5.5)	Significance
t50 (50% release)	23.2 h	44.4 h	Faster endo = better escape
t90 (90% release)	48.0 h	N/A	Complete release window
Rate constant k	0.0300 /h	0.0156 /h	Higher endo = pH-responsive
Max released	75%	75%	= encapsulation efficiency

5. Shelf-Life & Degradation Predictor

Grade: POOR (<3 months) | t90 = 77 days | Dominant pathway: Aggregation | Storage: 4 degC

Pathway	Rate (k, /day)	Relative %
Hydrolysis	0.000061	4%
Oxidation	0.000160	12%
Aggregation	0.000781	57%
Drug leakage	0.000368	27%

6. Colloidal Stability — DLVO Theory + Protein Corona

DLVO potential energy V_total = -6.2 kT (UNSTABLE ✗ — REFORMULATE) | Threshold: >25 kT prevents aggregation in blood.

Parameter	Value	Equation	Significance
DLVO V_total (kT)	-6.2	V_vdW + V_EDL	>25kT = stable
V_vdW (attraction)	N/A	-A·R/(12h)	Hamaker constant
V_EDL (repulsion)	N/A	$64\pi\ \varepsilon\ R\ \gamma^2\ e^{-\kappa h}$	Zeta potential
Debye length (nm)	N/A	$1/\kappa$	Ionic screening
Protein corona (nm)	N/A	Langmuir model	Stealth reduction
Opsonin index	N/A	IgG + C3b binding	<0.3 = good
Stealth index	1.000	PEG brush model	0.6+ = good
MPS clearance (h)	N/A	Michaelis-Menten	>12h adequate

7. Nanotoxicity & Immunogenicity Screening

Immunogenicity grade: **CAUTION** | Score: 30/100

Test	Score / Result	Risk	Threshold
CARPA (complement)	0.520	MODERATE	>0.6 = HIGH
Anti-PEG antibody	0.250	LOW	0.5+ = caution
MPS uptake	0.240	—	>0.6 = rapid clearance
Cytokine storm	—	LOW	Charge-dependent
Platelet activation	—	LOW	Cationic >200nm

8. Off-Target Toxicity — 50-Receptor QSAR Panel

Overall: CAUTION -- 4 high-risk targets | Cardiac risk: NO ✓ | Hepatic risk: NO ✓ | CNS off-target: YES ■

Receptor	Free Drug Score	In-DDS Score	Risk
hERG_K+	0.000	0.000	LOW
Nav1.5	0.000	0.000	LOW
Cav1.2	0.000	0.000	LOW
CYP3A4_inhib	0.305	0.191	LOW
CYP2D6_inhib	0.722	0.451	MODERATE
CYP2C9_inhib	0.357	0.223	LOW
CYP1A2_inhib	1.000	0.625	HIGH
CYP2C8_inhib	0.205	0.128	LOW
OATP1B1	0.303	0.189	LOW
MDR1_Pgp	0.000	0.000	LOW
BCRP	0.245	0.153	LOW
MRP2	0.093	0.058	LOW
DAT_dopamine	0.928	0.580	HIGH
SERT_serotonin	0.721	0.450	MODERATE
5HT2A	0.629	0.393	LOW
D2_dopamine	0.938	0.586	HIGH
GABA_A	0.242	0.151	LOW
NMDA	0.496	0.310	LOW
sigma1	1.000	0.625	HIGH
ERalpha_estrogen	0.000	0.000	LOW

High-risk flags (4):

■ CYP1A2_inhib: HIGH (score=0.62)

■ DAT_dopamine: HIGH (score=0.58)

■ D2_dopamine: HIGH (score=0.59)

■ sigma1: HIGH (score=0.62)

9. Advanced Science Modules — Key Outputs

9a. Competitive DDS Landscape (ClinicalTrials.gov)

Our DDS: BBB Score = 100 | Position: SUPERIOR | Active trials found: 4

NCT ID	Title	Phase	N
NCT04899908	Stereotactic Brain-directed Radiation With or	['PHASE2']	134
NCT06855368	Biomarkers for Monitoring Dementia Progressio	['?']	300
NCT06846658	Exploring the Olfactory Mucosa, Blood and Uri	['?']	180
NCT07667387	Study of LNP.UCD.ABE in Patients With Urea Cy	['PHASE1', 'PHASE2']	7

9b. Patient Subgroup Stratification

Overall response: 82.2% | Best: Very elderly (>80) (84%) | Worst: Young adults (18-40) (79%) | Recommendation: Highest efficacy in Very elderly (>80) (predicted response 84%). Dose reduce by

Subgroup	Pop. Freq.	CNS Bioavail	Response%	Tox Risk	Dose Adj.
Young adults (18-40)	20% of patients	36.7%	79.3%	LOW	x1.29 standard dose
Middle-aged (40-65)	35% of patients	41.8%	82.3%	LOW	x1.14 standard dose
Elderly (65-80)	30% of patients	42.7%	82.8%	LOW	x0.72 standard dose
Very elderly (>80)	15% of patients	46.2%	84.4%	MODERATE	x0.46 standard dose

9d. Lysosomal Trafficking Predictor

Cytosolic route: 35% | Lysosomal degradation: 63% | Nuclear entry: 2% | Concern: HIGH -- significant lysosomal degradation expected | Mitigation: Add proton sponge polymer (PEI or PAMAM) to improve endosomal escape

10. Synthetic Clinical Trial — N=500 Virtual Patients

Monte Carlo simulation of Phase 1 trial. Patient covariates: age, weight, CYP3A4 genotype, renal function, BBB integrity by age. Pharmacokinetics computed per-patient from PBPK model.

NO-GO / REFORMULATE	100.0%	72.6%	0.09 mg/kg	500
Go/No-Go	Response Rate	Severe AE	Optimal Dose	N Patients

Endpoint	Result	Benchmark	Status
Overall response rate	100.0%	>60% = GO	✓
Severe AE rate	72.6%	<5% = GO	✗
Young adults (<65)	100.0%	—	—
Elderly (>65)	100.0%	—	—
Mild AE	27.4%	—	—
Renal AE	26.8%	<15%	✗

Recommendation: Predicted Phase 1 success rate: 100%. Dose 0.09 mg/kg recommended. Monitor renal function in 27% of elderly patients.

11. Manufacturing & Sterilization Modules

11a. Terminal Sterilization Survivability

Recommended method: Gamma 25Kgy | Cost implication: Gamma or E-beam (~3x cost) | Feasible methods: gamma 25kGy, aseptic filtration 022um, VHP H2O2

Method	Survives?	Detail
autoclave 121C	NO ✗	Biologic MW=6kDa -- THERMAL DENATURATION
gamma 25kGy	YES ✓	Survives gamma sterilization
aseptic filtration 022um	YES ✓	Size 80 nm < 220 nm -- filterable
VHP H2O2	YES ✓	PEG provides surface protection

11b. Lyophilization Cycle Optimizer

Parameter	Value	Standard
Tg' (cryo.	-30°C	Formulation-specific
Primary drying T	-32.0°C	Must be < Tg'
Primary P	0.1478 mbar	0.5×P_ice
Primary time	25.6 h	Sublimation model
Secondary T	25.0°C	+25°C
Cycle total	39.6 h	Include ramp time
Cake collapse	OK: primary drying fixed at Tg'-2degC safety margin (-32 degC)	Must say OK

11c. Cryo-Chain Excursion Predictor

Excursion: -20.0°C for 4.0h | EE: 75.0% → 74.2% (loss: 0.8%) | Decision: RELEASE BATCH | Confidence: 98%

11d. Adversarial Stress-Testing

Grade: 4/5 scenarios passed -- CONDITIONAL | 4/5 scenarios passed.

Scenario	Pass/Fail	Detail
Fever 40C	FAIL ✗	Tm=42.0degC -- RISK: may partially melt at fever temperature
Shear Stress	PASS ✓	Carrier withstands capillary shear
Oxidative Stress	PASS ✓	Polymeric carrier -- relatively resistant to oxidative stress
High Protein Plasma	PASS ✓	Stealth index 1.00: adequate MPS evasion
Repeated Dosing	PASS ✓	Anti-PEG IgM may reduce t1/2 to 70% of first dose after re-dosing. Mon

12. Regulatory Compliance & Supply Chain

12a. FDA 21 CFR Part 11 Compliance

Status: NOT VERIFIED -- no audit trail entries found for this trial. log_computation() must be called during computation to build a compliant audit trail; this trial has none. | Audit entries: 0 | Data integrity: N/A -- no audit entries to hash-chain

12b. Geopolitical Supply-Chain Risk Assessment

Overall supply risk: HIGH | Supply chain score: 0/100 | Recommendation: Establish 6-month buffer stock for high-risk materials

Material	Risk	Source	HHI Index	Mitigation
ApoE3 peptide	VERY HIGH	Custom synthesis	0.85	Qualify alternative supplier

12c. Digital Pharmacovigilance — Post-Elimination Fate

Unchanged excretion: 10.0% | Eco-persistence: 1.0 days | Environmental risk: LOW | Renal dose adjustment: No adjustment

13. Literature Citations — PubMed Auto-Mined

Top relevant publications retrieved live from PubMed E-utilities API based on drug name + carrier type. These papers confirm or contextualize the computational predictions above.

[1]

[2]

14. Biophysical Binding & CNS-Specific Modules

14a. FEP+ Binding Affinity (LIE approximation)

Ligand: apoe | ΔG_single = -5.00 kcal/mol | ΔG_avidity = -2.59 kcal/mol | Kd = 14975652 nM (Weak (>100nM)) | n_ligands = 16085 | Residence time = 66.8 s

14b. Glymphatic Clearance Simulation

ECM binding index = 0.480 | t½ waking = 9.0 h | t90 clearance = 30.0 h | CAUTION: small particles cleared rapidly by glymphatics

14c. Microglial Activation & Neuroinflammation

Neuroinflammation score = 0.020 | Risk: LOW -- safe for CNS application | IL-6 fold-change = 1.2× | TNF-α = 1.2×

→ No mitigation needed

14d. FUS-Responsive Nanocarrier Design

Frequency = 0.5 MHz | MI target = 0.40 | BBB open window = 42 min | FUS uptake = 68.0% | Inject 80 nm DDS 5 min before FUS. FUS opens BBB for 42 min. Predicted 68% enhan

14e. 62-Principle C+ Flow — Class A Surrogate (Top-1 DDS)

Top-1 DDS: ApoE-LNP **Composite**: 78.1/100 **Verdict**: GOOD

ApoE-LNP: composite CNS-principle score = 78.1/100 (GOOD). Carrier: lnp, size 0.0 nm, ζ +0.0 mV, ligand: ApoE. Strongest group: G4 Safety (94.1/100); weakest: G8 Translational (0.0/100). Evaluated against all 57 Class A surrogate principles.

CNS Principle Group Rollups

CNS Group	Score (0-100)
G1 CNS Delivery	60.6
G2 Release Kinetics	68.8
G3 Stability	63.0
G4 Safety	94.1
G5 Glymphatic BBB	60.0
G6 Manufacturability	65.9
G7 DrugDDS Fit	63.7
G8 Translational	0.0

Top Strengths (Class A Surrogate)

Principle	Score	Method	Reference
P05	100.0	% subgroups responding (lower CYP risk = more univ	
P09	100.0	100 - sum(metabolite organ accumulation risks)	
P10	100.0	Non-LNP carrier — falls back to drug-side ionizati	

Weak Spots (Below 60)

Principle	Score	Method	Reference
P43	25.0	FUS-response by carrier acoustic properties	
P60	15.0	Swarm capability (most carriers: no)	
P44	1.0	AUC over 24h therapeutic window proxy	

14e-ii. Complete 62-Principle Scoreboard (P01 → P62)

All 62 principles in canonical order, covering Class A (Surrogate), Class B (Deep Physics), and Class C (Translational).

P#	Cls	Title	Score	Conf.	Method
P01	B	Adversarial Stress-Testing Engine	57.5	MODERATE	Mean of 6 stress scenarios (pH4.5, pH2, pH8.5, 42°
P02	B	Cross-Species PK Scaling (Allometric)	75.0	MODERATE	BW^0.75 scaling + class-specific adjustment
P03	A	Competitive DDS Landscape Radar	92.0	LOW	Novelty = 100 - frequency_of_combo_in_CNS_trials,
P04	B	Quantum Coherence Transport Model	27.1	MODERATE	WKB tunneling: P ∝ exp(-2·barrier)
P05	A	In-Silico Patient Subgroup Stratifier	100.0	MODERATE	% subgroups responding (lower CYP risk = more univ
P06	A	Lysosomal Trafficking Predictor	79.3	MODERATE	Carrier endosomal_escape ⊕ drug-passive permeation
P07	A	Real-Time Literature Mining (PubMed)	80.0	LOW	log-count proxy of literature support
P08	B	Degradation Kinetics Under Oxidative S	43.0	MODERATE	Arrhenius k=A·exp(-Ea/RT) for carrier; minus penal
P09	A	Digital Pharmacovigilance Engine	100.0	MODERATE	100 - sum(metabolite organ accumulation risks)
P10	B	LNP Ionization State Predictor	100.0	LOW	Non-LNP carrier — falls back to drug-side ionizati
P11	B	Formulation Instability Fingerprint	100.0	MODERATE	100 - sum(weak-bond penalties from SMILES)
P12	B	CNS Disease-Stage-Aware Dosing	91.5	MODERATE	Score = 100·BBB_integrity_factor × (0.4 + 0.6·CNS-
P13	B	PBPK Digital Twin	0.7	MODERATE	AUC_brain/AUC_plasma proxy; 3-compartment surrogat
P14	A	In-silico Dissolution & Release Profil	100.0	MODERATE	t50 from carrier kinetics × drug-membrane partitio
P15	A	Shelf-life & Degradation Predictor	46.2	MODERATE	Arrhenius baseline × (0.7 + 0.3·EE) × drug_hydroly
P16	B	Scale-up & Manufacturability	48.0	MODERATE	Scale-readiness × shear-robustness, minus drug-pro
P17	A	Nanotoxicity & Immunogenicity Screenin	91.2	MODERATE	Mean of (hemolysis penalized by drug net+ charge,
P18	B	Active Targeting & Receptor Binding	85.4	HIGH	Ligand-affinity table × density × drug-compatibili
P19	A	QbD - Quality by Design Engine	100.0	MODERATE	Fraction of CQAs (size, zeta, EE, PDI, drug-loadin
P20	A	Cost-Efficiency Engine	62.8	LOW	Score = 100 - 3·($/mg estimate, drug+carrier+ligan
P21	C	Pre-IND Regulatory Reports	0.0	—	skipped_deep_validation_insufficient
P22	A	Protein Corona Predictor	83.3	MODERATE	100 × exp(-corona_thickness/10); thickness from ca
P23	B	Dynamic Crystal Polymorphism	80.0	MODERATE	100 - polymorph risk (rot_bonds + H-bond pattern)
P24	B	Shear-Stress & Scale-Up Collapse	59.6	MODERATE	log10(crit_shear/operating_shear) × 30; crit_shear
P25	A	Extractables & Leachables	100.0	MODERATE	100 - leachables risk by drug LogP × carrier

P#	Cls	Title	Score	Conf.	Method
P26	A	Microbiome-Excipient Interactions	78.0	MODERATE	100 - mean_microbiome_degradability(carrier) - dru
P27	A	Lyophilization Cycle Optimizer	62.5	MODERATE	100 - (|Tg'| - 5) × 1.5
P28	A	3D-Printed Polypill Rheology	50.0	MODERATE	Carrier-specific printability index
P29	B	Biomimetic & Exosome Engineering	65.0	MODERATE	Stealth-from-macrophage score; PEG ≥5% boost
P30	B	QM/MM Stimuli-Responsive Cleavage	45.0	MODERATE	Score = 50 × |7.4 - pH_trigger|
P31	B	In-Silico Biodistribution	0.4	MODERATE	brain% = BBB · ligand · size_match · 100; 100 if ≥
P32	C	Automated FTO & IP Evader	0.0	—	skipped_deep_validation_insufficient
P33	B	BBB Quantum Breaker (Trojan-Horse Desi	54.6	MODERATE	(0.6·ligand + 0.4·size) × density_factor
P34	A	DNA Logic Gates & Bio-computing	25.0	MODERATE	DNA-based carriers score high
P35	A	Microgravity Formulation Engine	100.0	LOW	Sedimentation-Péclet proxy; smaller = better
P36	A	Geopolitical Supply-Chain Resilience	60.0	MODERATE	Supplier-diversity index by carrier + ligand
P37	A	Eco-Destructible Pharma	60.0	MODERATE	Biodegradability index by carrier class
P38	B	Glymphatic Clearance Trap	100.0	MODERATE	Stokes-Einstein: optimum 80-150 nm
P39	A	Microglial Activation & Neuroinflammat	100.0	MODERATE	100 - (cationic_charge + carrier_risk - PEG_protec
P40	B	Intranasal-to-Brain Delivery	55.0	MODERATE	Mucoadhesion + thermo-responsive boost
P41	B	Exosome Cargo Loading Thermodynamics	40.0	LOW	Not an exosome carrier — N/A
P42	A	Region-Specific Spatiotemporal Navigat	75.0	MODERATE	Ligand-region specificity table
P43	B	FUS-Responsive Nanocarriers	25.0	MODERATE	FUS-response by carrier acoustic properties
P44	B	CNS-Specific PBPK Time-Machine	1.0	MODERATE	AUC over 24h therapeutic window proxy
P45	C	FDA 21 CFR Part 11 Compliance	0.0	—	skipped_deep_validation_insufficient
P46	A	Polypharmacy & DDI Simulator	100.0	MODERATE	CYP-inhibition heuristic from SMILES
P47	B	Free Energy Perturbation (FEP+)	90.0	LOW	Vina-like ΔG proxy from MW + LogP (deep mode runs
P48	A	Off-Target Toxicity & QSAR (50-recepto	100.0	MODERATE	50-receptor QSAR surrogate (hERG, 5HT2B, AhR risks
P49	A	Organ-on-a-Chip Simulator	80.0	MODERATE	Microfluidic compatibility (size + PDI)
P50	B	Cryo-Chain Thermal Excursion Predictor	71.5	MODERATE	Distance from -20°C phase transition × 1.3
P51	B	Terminal Sterilization Survivability	60.0	MODERATE	Carrier gamma-radiation survival (25 kGy)

P#	Cls	Title	Score	Conf.	Method
P52	A	Continuous Manufacturing Digital Twin	60.0	MODERATE	Continuous-process readiness by carrier
P53	A	Dark Data & Negative Results Vault	100.0	MODERATE	Similarity-to-documented-failure
P54	A	Pharmacogenomic-Guided Targeting	75.0	MODERATE	Class-based pharmacogenomic applicability
P55	C	Automated Grant & NIH Proposal Generat	0.0	—	skipped_deep_validation_insufficient
P56	C	Patentability Score Engine	0.0	—	skipped_deep_validation_insufficient
P57	B	Microfluidics & LNP Synthesis Digital	50.0	MODERATE	Microfluidic readiness for size 50-150 nm
P58	B	Impurity Cascade Predictor	80.0	MODERATE	Carrier residual-metals impurity profile
P59	B	4D Shape-Shifting Carriers	30.0	MODERATE	Stimuli-responsive bonus by release kinetics
P60	A	Swarm Nanorobotics Intelligence	15.0	LOW	Swarm capability (most carriers: no)
P61	B	Synthetic Clinical Trials & Virtual Hu	30.1	MODERATE	Virtual-cohort responder fraction
P62	A	Biobetter / Supergeneric Generator	65.0	LOW	Novelty distance from on-market CNS DDS

62 principles evaluated. Composite: 78.1/100 (GOOD)

14f. Class B Deep Physics Validation (Top-1 DDS)

Verdict: FAILED **Pass rate**: 39.3% (11/28 principles validated, threshold 70%)

Combined score (surrogate pass-through + real physics): 11/28 principles scored PASSED (39.3%). Of these, only 7/28 principles ran independent deep computation (passed 3/7 = 42.9%); the rest re-used their Class-A surrogate score pending a future full-physics HPC run. Cite independent_pct, not pct, as evidence of physics-based validation.

Per-principle deep validation

P#	Validated	Score	Value	Conf.	Method
P01	✗	57.5	57.55	LOW	Full MD stress simulation — surrogate value confirmed (
P02	✓	75.0	3744.0	MODERATE	Resolved clinical half-life (bundle tier 1); Mahmood 20
P04	✗	27.1	0.1353	LOW	Full QM tunneling (QCElemental + ASE) — surrogate value
P08	✗	43.0	43.0	LOW	Full radical-chain reaction MD — surrogate value confir
P10	✓	100.0	100	MODERATE	Constant-pH MD simulation — surrogate value confirmed (
P11	✓	100.0	100	MODERATE	DFT bond dissociation energies — surrogate value confir
P12	✓	91.5	91.5	MODERATE	Stage-specific PBPK with full BBB physiology — surrogat
P13	✗	4.2	0.0021	MODERATE	3-compartment PBPK ODE (scipy.integrate.odeint): blood
P16	✗	48.0	48.0	LOW	CFD of 1000L bioreactor — surrogate value confirmed (fu
P18	✓	92.0	-9.2	HIGH	MM/GBSA-style ΔG estimate from validated ligand-recepto
P23	✓	80.0	80.0	MODERATE	CSP via DFT — surrogate value confirmed (full-physics H
P24	✗	59.6	59.64	LOW	Full CFD + MD coupling — surrogate value confirmed (ful
P29	✓	65.0	65	MODERATE	Full membrane-fusion MD — surrogate value confirmed (fu
P30	✗	45.0	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confirmed
P31	✗	40.0	0.02	MODERATE	7-organ whole-body distribution with size + charge + li
P33	✗	54.6	54.6	LOW	Atomistic docking + SMD pulling — surrogate value confi
P38	✓	96.1	9.6	HIGH	Stokes-Einstein D = k_B·T/(6πηr) at 37°C, η_CSF = 7×10■
P40	✗	55.0	55	LOW	Full nasal cavity CFD — surrogate value confirmed (full
P41	✗	40.0	40	LOW	Full membrane mechanics MD — surrogate value confirmed
P43	✗	25.0	25	LOW	Full acoustic radiation force MD — surrogate value conf
P44	✗	0.0	0.0	MODERATE	4-compartment CNS-PBPK ODE with glymphatic clearance: B
P47	✗	73.8	-6.71	LOW — LI	LIE approximation (fallback — Vina unavailable)
P50	✓	71.5	71.5	MODERATE	Full coarse-grained MD lipid phase transition — surroga
P51	✓	60.0	60	MODERATE	Full radical-chain damage MD — surrogate value confirme
P57	✗	50.0	50	LOW	Full CFD of mixer geometry — surrogate value confirmed
P58	✓	80.0	80	MODERATE	Full QM/MM cascade simulation — surrogate value confirm
P59	✗	30.0	30	LOW	Full coarse-grained MD morphological transition — surro

P#	Validated	Score	Value	Conf.	Method
P61	✗	30.1	30.0	LOW	Full Monte-Carlo population PBPK — surrogate value conf

14g. Class C Translational Deliverables

Principle	Title	Status	Score / Outcome
P21		skipped_deep_validation_	—
P32		skipped_deep_validation_	—
P45		skipped_deep_validation_	—
P55		skipped_deep_validation_	—
P56		skipped_deep_validation_	—

Narratives

P21: Deep validation passed only 11/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P32: Deep validation passed only 11/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P45: Deep validation passed only 11/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P55: Deep validation passed only 11/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P56: Deep validation passed only 11/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

v23: structured outlines for Pre-IND, Grant, Patent will be rendered as fully formatted Word/PDF deliverables.

14h. Top-N Fallback Audit Trail

If the Top-1 DDS fails deep validation (≥70% threshold), the orchestrator falls back to Top-2, then Top-3. Each candidate's outcome and the explicit transition reasoning are recorded below.

Rank	DDS Name	Verdict	Pass %	Promoted?
#1	ApoE-LNP	FAILED	39.3%	—
#2	AAV9-vector	FAILED	32.1%	—
#3	RVG29-Liposome-pH	FAILED	42.9%	—

#1 ApoE-LNP — FAILED

Failure reason: Deep validation FAILED: only 11/28 principles validated (39.3%, threshold 70%). Critical failures: P01, P04, P08, P13, P16.

Transition reason: Falling back to rank #2 because deep physics showed insufficient evidence for this DDS.

Failed principles: P01, P04, P08, P13, P16, P24, P30, P31

#2 AAV9-vector — FAILED

Failure reason: Deep validation FAILED: only 9/28 principles validated (32.1%, threshold 70%). Critical failures: P01, P04, P08, P13, P16.

Transition reason: Falling back to rank #3 because deep physics showed insufficient evidence for this DDS.

Failed principles: P01, P04, P08, P13, P16, P18, P24, P30

#3 RVG29-Liposome-pH — FAILED

Failure reason: Deep validation FAILED: only 12/28 principles validated (42.9%, threshold 70%). Critical failures: P04, P08, P13, P16, P24.

Transition reason: No more candidates in Top-3 — reverting to rank #1 and reporting with FAILED verdict. Researcher should reformulate.

Failed principles: P04, P08, P13, P16, P24, P30, P31, P33

15. Executive Decision Framework

Criterion	Status	Evidence
DLVO colloidal stability	FAIL ✗	V_total = -6 kT
BBB penetration >10%	PASS ✓	DDS BBB = 100.0%
Cardiac off-target	PASS ✓	hERG/Nav/Cav QSAR
Clinical trial GO	NO-GO / REFORMULATE	Response 100% | AE 72.6%
Sterilization feasible	3 method(s)	Gamma 25Kgy
Supply chain	HIGH	Score 0/100
OVERALL DECISION	CONDITIONAL GO	Reformulate/re-evaluate before IND-enabling studies — failed: synthetic trial returned 'NO-GO / REFORMULATE'; colloidal instability (DLVO)

NOTE: All results are computational predictions. Required wet-lab validation: DLS/Zeta (physical stability), HPLC/NMR (drug content), BBB TEER assay, in vivo PK in rodent model.

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